

Induction Heads

The circuit that copies from context · video 2:11

This lesson answers one question: how does a transformer copy from its context? GPT-2 can't predict 50 random tokens (loss 12.8), but when they repeat, it predicts the second copy almost perfectly (loss 0.25). The work is done by **induction heads**: attention heads that find the earlier occurrence of the current token and look at what came next. Scoring all 144 heads finds five strong ones in layers 5–7; removing them destroys the copying, while removing random heads barely matters.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 6).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. Copying from context	▶ 0:08–0:47	1	☐
2. Finding the heads	▶ 0:47–1:39	3	☐
3. Why it matters	▶ 1:39–2:02	1	☐

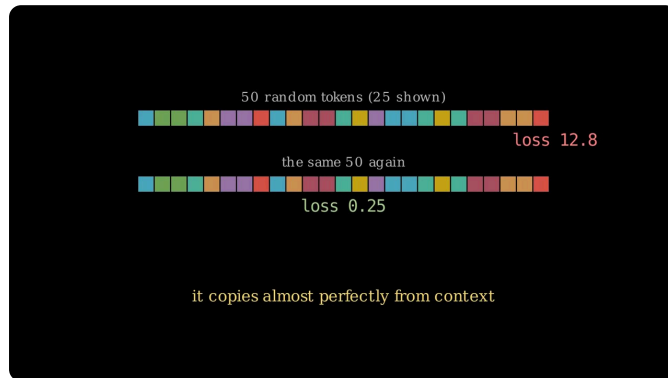
Before you start: Deep Dive episodes 8 (the attention matrix) and 13 (the residual stream). Code: `code/d42_induction_heads` (GPT-2 small).

CONCEPT 1 OF 3

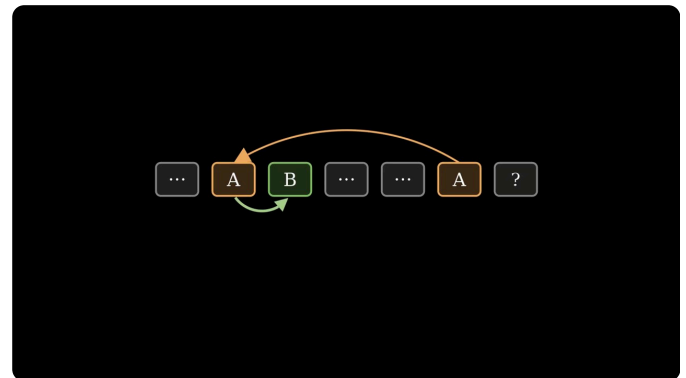
Copying from context

▶ Video 0:08–0:47

Random tokens are unpredictable: on 50 of them GPT-2's loss is **12.8**. Repeat the same 50 tokens, and on the second copy the loss is **0.25**. The model copies almost perfectly from its context. The rule it applies: the current token is A; find where A appeared before; look at the token right after it, B; predict B.



▶ 0:37 50 random tokens, then the same 50 again: loss 12.8 on the first copy, 0.25 on the second.



▶ 0:46 The induction rule: the current token is A; find the earlier A; predict the token that followed it, B.

KEY IDEA

Repeated random tokens isolate copying: nothing but the context can predict them.

Check 1: Copying from context

Answer all 1 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

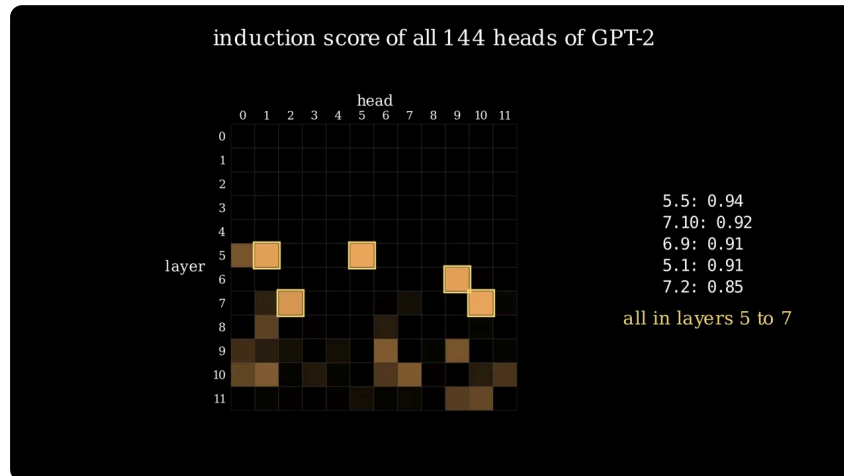
A loss of 0.25 means the model gives the right token what probability, on average (geometric mean)? And a loss of 12.8?

CONCEPT 2 OF 3

Finding the heads

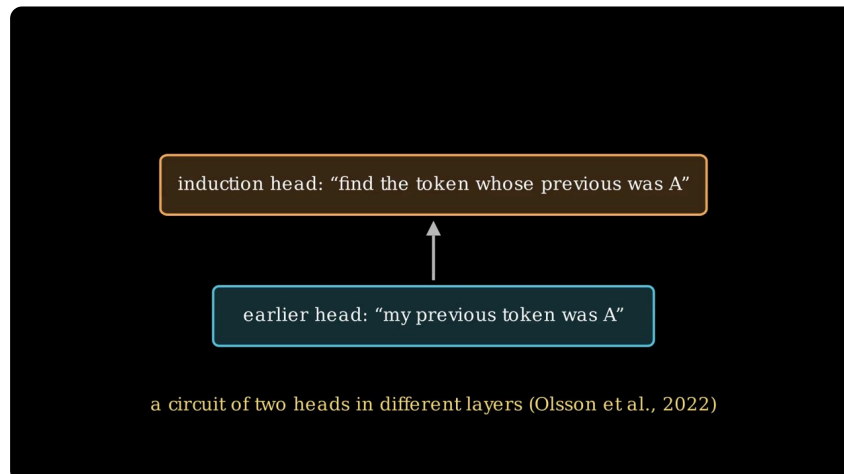
▶ Video 0:47-1:39

An induction head is an attention head that performs that lookup. Its **induction score**: in the second copy, the average attention from each token to the token right after its earlier occurrence. Of GPT-2's 144 heads most score near zero, 15 score above 0.3, and five score above 0.8: 5.5 (0.94), 7.10 (0.92), 6.9 (0.91), 5.1 (0.91) and 7.2 (0.85).



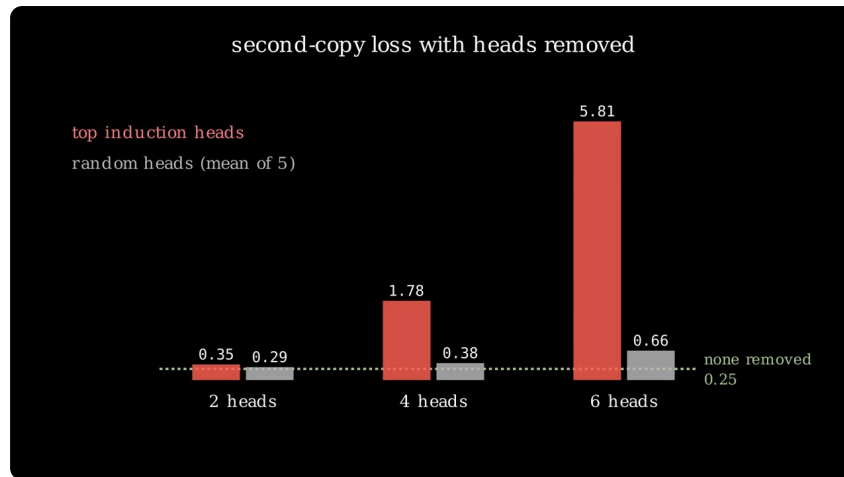
▶ 1:02 Induction score of all 144 heads: five above 0.8, all in layers 5-7 (5.5, 7.10, 6.9, 5.1, 7.2).

Why not in layer 0? To find “the token after A”, each position must first know what its previous token was. In the original analysis, an earlier **previous-token head** writes that into the residual stream and the induction head reads it: a circuit of two heads in different layers. (On these random sequences no single early GPT-2 head scores strongly for attending to the previous token, so that half is spread out or done differently here.)



▶ 1:18 The circuit: an earlier head records each token's predecessor; the induction head searches for it (Olsson et al., 2022).

Removing heads (head_mask = 0) tests their role. The top 2 induction heads: second-copy loss 0.25 → 0.35; the top 4: 1.78; the top 6: 5.81. Removing 6 random heads instead: 0.66 on average.



▶ 1:37 Second-copy loss with heads removed: top 2/4/6 induction heads 0.35/1.78/5.81; random heads 0.29/0.38/0.66.

KEY IDEA

Score heads by a behaviour, then ablate them: a high score plus a large ablation effect identifies a mechanism.

Check 2: Finding the heads

Answer all 3 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

In the second copy, position t (counting from 0, with $T = 50$ tokens per copy) holds the same token as position $t - 50$. Which position should an induction head attend to?

2.2 MULTIPLE CHOICE

Why compare removing the top induction heads with removing random heads?

- A. To speed up the experiment
- B. To show the effect comes from those heads, not from removing any 6 heads
- C. Because random heads are induction heads
- D. To measure the first copy

2.3 HANDS-ON CODE

Try it yourself. Compute one head's induction score.

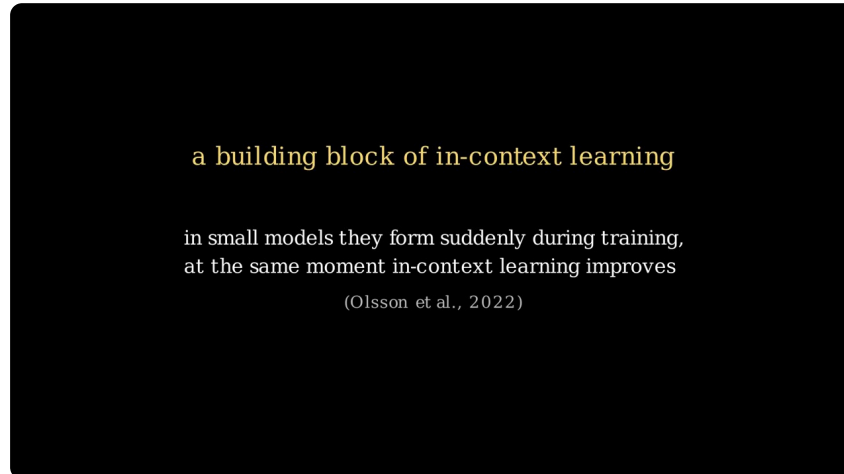
```
import torch
from transformers import GPT2LMHeadModel
m = GPT2LMHeadModel.from_pretrained("gpt2", attn_implementation="eager")
seq = torch.randint(1000, 30000, (4, 50)); ids = torch.cat([seq, seq], 1)
a = m(ids, output_attentions=True).attentions[5][:, 5]
print(sum(a[:, t, t - 49].mean() for t in range(50, 100)) / 50)
```

CONCEPT 3 OF 3

Why it matters

▶ Video 1:39–2:02

Induction heads are a building block of **in-context learning**: using patterns from earlier in the prompt. Research on small models (Olsson et al., 2022) found that they form suddenly during training, at the same moment in-context learning improves. In code, the score is a single average over attention weights.



▶ 1:51 Induction heads: a building block of in-context learning.

KEY IDEA

Some model behaviours are carried by small, findable circuits.

Check 3: Why it matters

Answer all 1 questions before looking at the answers. Pass mark: all correct.

3.1 TRUE OR FALSE

“An induction head can copy a token sequence it never saw during training.”

True ☐ False ☐

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. Copying from context · rewatch ▶ 0:08-0:47

1.1 About 0.78; about 0.000003. $e^{-0.25} \approx 0.78$; $e^{-12.8} \approx 2.8 \times 10^{-6}$.

2. Finding the heads · rewatch ▶ 0:47-1:39

2.1 $t - 49$. The token after the earlier occurrence: $(t - 50) + 1$.

2.2 B. Any ablation hurts a little; the control shows how much is specific.

2.3 About 0.9 (head 5.5). Head 5.5 scored 0.94 in the episode's run.

3. Why it matters · rewatch ▶ 1:39-2:02

3.1 True. The sequences here are random; the head copies whatever the context contains.